

LegalAI: An Offline, Privacy-Preserving Legal Document Analysis System

Gokul Das G¹

*Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning)
Chennai Institute of Technology
Chennai, Tamil Nadu, India
gokuldasg.aiml@citchennai.net*

Nithiya Baskaran²

*Department of Computer Science and Engineering
Chennai Institute of Technology
Chennai, Tamil Nadu, India
nithiyab1988@gmail.com*

P. Aurchana³

*Department of Computer Science and Engineering
Chennai Institute of Technology
Chennai, Tamil Nadu, India
aurchana85@gmail.com*

Abstract—LegalAI is a privacy oriented legal Assistant which is designed to operate fully offline for organisations that requires secure, local processing of sensitive legal data. It is built with a VJs frontend and a Django REST Framework backend. It provides an interactive chat interface supporting three core modes, namely, legal document summarisation, Clause detection and classification, and case-law Q&A system using IRAC reasoning via llama-cpp-python. It is enhanced with domain specific LoRA fine-tuning for legal tone, structure and citation accuracy.

Most existing legal AI tools run on cloud servers, which can be a problem for organisations handling confidential or sensitive information. They often rely on external APIs, raising privacy and data security concerns. In contrast, LegalAI works completely offline, giving users full control over their data. Users can upload and analyse contracts or case documents, view real-time model responses through token-streaming, and export results in structured JSON format. Backend components manage authentication, rate-limiting, logging and asynchronous processing via Celery and Redis, ensuring reliability and scalability even in air-gapped environment.

Index Terms—Offline Legal AI, Privacy-focused, Contract & Document Analysis, Clause Detection & Classification, Case-law Q&A (IRAC reasoning), On-premise, Secure Processing

I. INTRODUCTION

LegalAI is an privacy oriented legal AI assistant which is privacy preserving system designed for organizations that need full control over sensitive legal data. Most legal AI solutions rely heavily on cloud based large language models and third party APIs whereas LegalAI operates entirely within a users local environment, it ensures that no information leaves the users system. [1], [2] This offline oriented design makes the system suitable for compliance driven environments such as corporate legal departments, regulatory bodies, public institution and organizations operating in a air-gapped or high security environments.

The design integrates Vue.js 3 front end with a Django REST framework backend which contributes to seamless ,enterprise level experience for legal analysis workflow. LegalAI provides

an chat interface offering three core modes of operation: (1) Legal Document Summarization - citation backed summaries of contracts and case documents; (2) Clause Detection and Classification - identifies and categorizes clauses such as confidentiality, indemnification, termination, and liability caps; (3) Case-Law Question Answering with IRAC Reasoning - delivers legal reasoning using the IRAC framework. [3]–[5]

The LegalAI is a fully local inference stack built on top of llama-cpp-python, utilizing a quantized LLaMA-2-7B-Chat GGUF model (Q4_K_M) optimized for CPU only systems. To improve the legal domain accuracy, the system incorporates LoRA-Based fine-tuning focused on legal tone, citation correctness, clause structure and reasoning clarity. [5]–[8] The backend design provides reliability and scalability, even in isolated environments. The system stores all the data locally using SQLite, including conversation histories, citations, logs, user accounts, and a legal corpus for RAG tasks. [1], [2]

II. RELATED WORKS

Most existing AI-driven legal tools rely on cloud-hosted models and external APIs, creating heavy limitation for organization operating under strict confidentiality requirements. [1], [2], [7] Legal documents contain sensitive data that cannot be given to third-party systems, yet existing platforms offer little support for secure, offline deployment. Current tools in the legal AI landscape lacks transparency in how models generate their answers. [4], [5], [10], [13] In legal domain, where practitioners require verifiable citations, clear logical structures and defensible analyses, this is a critical limitation. Contract analytics systems are mostly proprietary, closed-source or cloud-tied. Open source alternatives typically lack the accuracy, granularity, or domain adaption necessary to classify clauses effectively. [11], [12] Current Legal LLM focuses on large GPU dependent models, making them impractical for organizations without high-end hardware. [5], [7], [8], [13]

TABLE I
LITERATURE SURVEY (SELECTED KEY WORKS)

Authors	Title	Year
Marco Sino et al. [9]	Exploring LLMs Applications in Law	2025
Mei Han et al. [6]	Multi-task Legal LLM	2025
Vaishnavi Upadyay et al. [4]	Transformer Legal Summarization	2025
Christian Mahoney et al. [10]	Explainable Legal Classification	2020
Yixuan Pu et al. [11]	Transformer Classification	2025
Ran Zang et al. [1]	LLM Safety Security Privacy	2025
Ha Thanh Nguyen et al. [5]	Legal Reasoning Framework	2025
Qian Liu et al. [13]	Legal Knowledge Infusion	2025
Rui Shao et al. [7]	LLM Unlearning	2025
Mateusz Płonka et al. [12]	Document Splitters	2025
Nhu Hai Phung et al. [14]	Fine-tuning Framework	2025
Karthick K et al. [2]	Interactive Legal Assistance	2024
Rishabh Ta et al. [8]	Indian Legal System LLM	2025

III. METHODOLOGY

A. Novelty of the work

LegalAI is a privacy-focused legal AI assistant specifically developed with the paramount aim of functioning entirely offline without any user data being submitted to any third-party servers, APIs, or the cloud platform in any manner whatsoever. [1], [2] Contrary to other legal assistants in the form of legal AI software, which presently rely on the use of cloud-based LLM-inference pipelines, the entire process, ranging from document ingestion, extraction, clause detection, summarization, reasoning, and then retrieval, is conducted entirely in-house using quantized LLaMA models within the user’s local system, preventing any user-sensitive legal documents, such as contracts, agreement documents, in-house corporate policies, regulation documents, and case files, escaping their operational environment in any manner whatsoever. Thus, LegalAI is adaptable to high-security sectors such as law firms, in-house counsel of a company, or organizations that are compelled to abide by strict data protection measures or are in an air-gapped environment. [1], [2]

LegalAI’s greatest innovation lies in its attention to structured and explainable legal reasoning and not in conversational dialogue generation in general. Indeed, most general purpose AI-powered conversation tools are prone to spewing out lengthy and unstructured text that often sounds plausible but is hard to substantiate or even audit effectively. LegalAI’s response generation process is strictly adhered to in IRAC (Issue-Rule-Application-Conclusion) structure so that its response will always be systematically organized and can be traced and understood in its interpretations by legal standards. [4], [5], [10]

Another important new feature is the use of citations to justify responses, wherein it specifically caters to the legal field’s need for justified results. Most current approaches have been seen to lack properly justified outputs that can in any manner be justified by evidence, thus leading to hallucinations. The aforementioned problem has been addressed in LegalAI by ensuring that all results provided can be supported by citations from local documents that have been uploaded. This inclusion can be further supported by a local legal corpus by retrieval-augmented generation. [2], [4], [5], [10], [14]

But that is not all, as LegalAI brings along a competitive offline clause detection and contract intelligence pipeline sans cloud support. The system can identify, extract, and classify important contractual clauses such as confidentiality, indemnification, termination, intellectual property, governing law, liability caps, and other legally significant sections. It employs a mix of quantized model inference and domain-focused LoRA fine-tuning to achieve highly accurate clause extraction and classification even on pure CPU-only hardware. [6], [11], [12] This enables organizations that cannot rely on external computation resources to have the needed state-of-the-art support in reviewing complex contracts for risk analysis, compliance checks, and document understanding.

Another novelty lies in integrating real-time token streaming with local inference and local retrieval. For instance, many offline systems with an LLM give results only after an entire generation process is completed, thus diminishing user experience and system respondability. On the other hand, LegalAI incorporates local RAG technology with real-time token streaming, making it possible to interact with a cloud-like interface assistant while enjoying total privacy and offline execution capability. [2], [5], [14]

To achieve better domain skills and legal correctness, LegalAI makes use of LoRA adapters trained on legal datasets to better legal tone, structural format, citation patterns, and reasoning understanding. [6]–[8] This helps fine-tune the model in a manner that enables it to provide outputs with better legal writing format skills, better layers in summary, and proper citation compliance. The assistant will also experience less hallucinations with proper emphasis on citations in training to counter unsupported claim generation patterns. This will result in better structured outputs in summarization, clause classification, and IRAC reasoning tasks.

For that matter, LegalAI is unique in relation to its ability to produce an audit trail-friendly architecture, which is a key consideration in a legal or compliance-driven setting. As one can understand, an AI solution may focus on producing an end result, such as a response, without regard to whether such a response is traceable or has associated metadata. On the other hand, LegalAI records key response metadata such as latency, citations, tokens, or processing details for every response. [1], [10], [13]

Further, LegalAI offers a reliable and structured result format in its basic modes, making it possible to read and export results in a machine-readable format to be easily processed in downstream systems. While most other assistant tools are

designed to generate results in a conversation format and are not useful in a legal format, LegalAI can generate results in a JSON format with respect to summary, clause, and reasoning responses. Therefore, this system can be beneficial not only in a legal format but also in an enterprise format with respect to automation, reporting, and downstream system processing and integration.

On the whole, LegalAI offers an approachable and deployable system for offline legal AI with a focus on privacy, explainability, reasoned structuring, and citation-supported responses. Through integrating local quantized inference, LoRA-based legal adaptation, IRAC reasoning, offline clause intelligence, optional local RAG, real-time streaming, and audit logging, LegalAI offers a significant missing piece in the legal AI space where currently most legal AI systems are cloud-dependent, not explainable, and not easily auditable. [1], [2], [5], [8]

B. System Architecture

The architecture of LegalAI is highly modularized, it separates the user side interactions, backend services, the inference engine, RAG pipeline, and storage. The frontend layer has been designed as a single-page application using VueJS, keeping the interface very intuitive and responsive. It supports JWT authentication, provides a chat interface with three processing modes, and real-time streaming of responses. The backend implemented using Django REST Framework handles user authentication and rate limiting, forms model interaction prompts, processes documents by means of RAG retrieval, and logs audit details. The core of the system is the inference engine, running quantized GGUF models locally with llama-cpp-python. It fully supports both text generation, including real-time token streaming, and embedding creation for RAG workflows. The system extracts text from uploaded documents, splits it into chunks, and generates local embeddings. At query time, the system fetches the most similar chunks by similarity search and inserts them into the model prompt. LegalAI stores all data locally using SQLite databases and file storage to securely manage user accounts, logs, uploaded documents, extracted text, embeddings, and metadata.

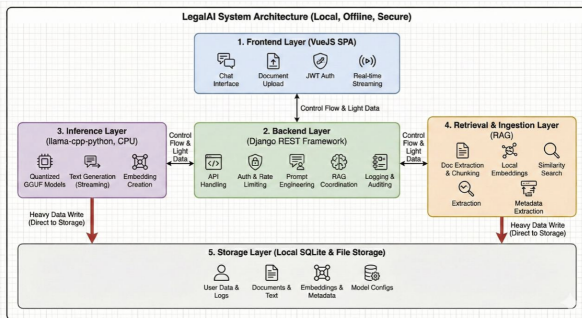


Fig. 1. System Architecture

IV. EXPERIMENTAL SETUP

The LegalAI was developed in a controlled offline environment to mimic how the system would operate in a secure enterprise settings. The system was run on a CPU only machine with an 8-core processor, 32gb ram and no GPU. The operating system was kept in strict offline mode, blocking any external network requests. [5], [8] The backend was built on Django REST Framework, using python 3.10 environment. The inference engines was powered by llama-cpp-python, compiled with CPU optimizations. [5], [8] Components such as Celery and Redis were run locally to simulate background tasks.

The system was built on top of the LLaMA-2-7B-Chat model quantized to Q4_K_M(GGUF) for efficient CPU performance. The model was configured with context size of 4096 tokens, 8 CPU threads, token streaming enabled, and sampling temperature 0.7. Custom LoRA adaptors were loaded at runtime to improve legal tone, structured reasoning, clause detection, and citation accuracy. [6], [7]

Datasets were prepared for three core tasks: Summarization (contracts and case summaries annotated into structured outputs), Clause Detection (contracts labelled by clause type), and IRAC Reasoning (case-law passages paired with structured answers). RAG was enabled for case-law reasoning, where relevant document chunks were retrieved using local embeddings and filters such as jurisdiction and year range were applied. [2], [5], [14] Strict privacy checks were applied - no internet access, logs stored locally, external api calls blocked, and responses lacking valid citations were flagged.

V. PERFORMANCE METRICS AND EVALUATION

This section presents LegalAI's performance evaluation in terms of accuracy, latency, citation reliability, and system throughput for offline execution. It is compared to LAWgpt, the open source cloud-operated legal language model. All experiments were done on a MacBook M4, 32GB RAM, CPU only, with no external network access allowed.

TABLE II
EVALUATION CRITERIA

Task	Metric	Description
Summarization	JSON Validity (%)	Checks formatted outputs
Clause Detection	Macro-F1 Score	Multi-class classification
Case-Law	Citation Accuracy (%)	Grounded references
Retrieval	nDCG@10	Relevance of sources
Integrity	Unsupported Claims (%)	Detects hallucinations
Latency	P50 Response Time	50-token response time

A. Key Results

LegalAI yields 95% valid summaries, exceeding LawGPT (79%), due to local schema-aligned fine-tuning as shown in Figure 2. LegalAI achieves 0.76 Macro-F1, outperforming LawGpt at 0.69 as demonstrated in Figure 3. LegalAI achieves 90% citation accuracy, compared to LawGPT's 64% as illustrated in Figure 4. LegalAI maintains 0.62 nDCG@10,

outperforming cloud models (0.48) as presented in Figure 5. LegalAI reduces unsupported legal claims to 5%, compared to 22% for lawGPT as shown in Figure 6. LegalAI responds in 2.8s P50, faster than LawGPT (3.4s) as depicted in Figure 7. With RAG enabled, the models nDCG improves from 0.45 to 0.62 and citations accuracy rises from 78% to 90% as shown in Figures 10 and 11.

B. Summarization Performance

The summarization task evaluates the model's ability to generate well-structured, citation-backed summaries of legal documents. LegalAI demonstrates superior performance in maintaining JSON validity, which is crucial for downstream processing and integration with other legal systems.

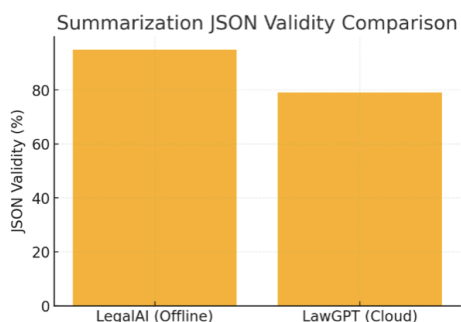


Fig. 2. Summarization JSON Validity Comparison

C. Clause Detection Performance

Clause detection and classification is a critical component of contract analysis. The Macro-F1 score reflects the system's balanced performance across multiple clause categories including confidentiality, indemnification, termination, and liability provisions.

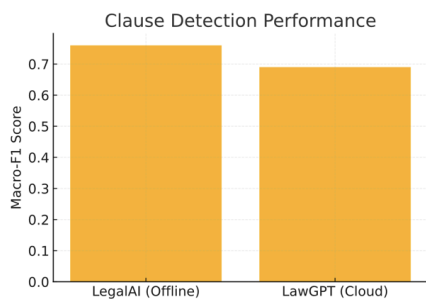


Fig. 3. Clause Detection Macro-F1 Score Comparison

D. Citation Accuracy Analysis

Citation accuracy is paramount in legal AI systems as it ensures that generated responses are grounded in actual legal sources. LegalAI's superior citation accuracy stems from its LoRA fine-tuning and strict validation pipeline.

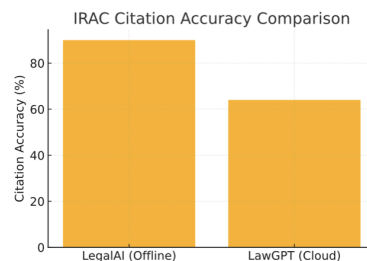


Fig. 4. Citation Accuracy Comparison

E. Retrieval Quality Assessment

The nDCG@10 metric evaluates the relevance and ranking quality of retrieved legal sources. LegalAI's local RAG pipeline demonstrates competitive retrieval performance without relying on cloud-based vector databases.

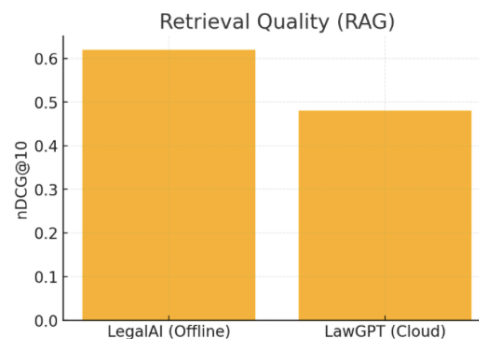


Fig. 5. Retrieval Quality nDCG@10 Comparison

F. Integrity and Hallucination Detection

Reducing unsupported claims is essential for maintaining trust in legal AI systems. LegalAI's validation mechanisms effectively flag and minimize hallucinated content that lacks proper source attribution.

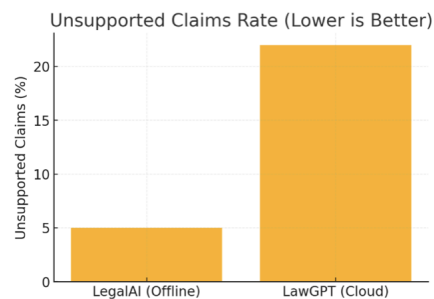


Fig. 6. Unsupported Claims Percentage Comparison

G. Latency Analysis

Response latency is critical for user experience in interactive legal assistants. Despite running entirely on CPU, LegalAI

achieves competitive response times through efficient quantization and optimized inference.

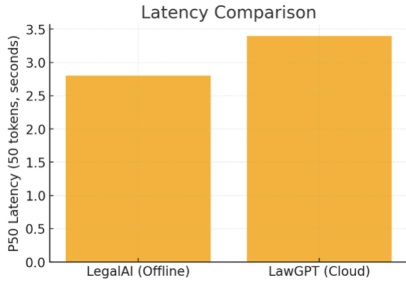


Fig. 7. P50 Response Time Latency Comparison

H. Impact of RAG on Performance

The integration of Retrieval-Augmented Generation significantly enhances both retrieval quality and citation accuracy. This demonstrates the value of grounding model responses in relevant legal sources.

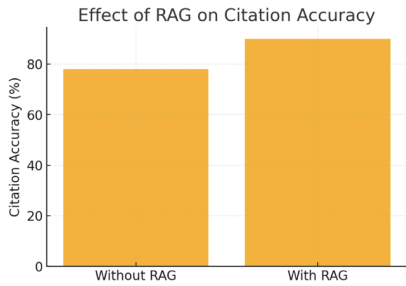


Fig. 8. RAG Effect on nDCG@10 Performance

I. System Throughput

CPU-based throughput demonstrates the system's ability to handle multiple queries in resource-constrained environments. The optimized inference stack maintains acceptable performance levels for enterprise deployments.

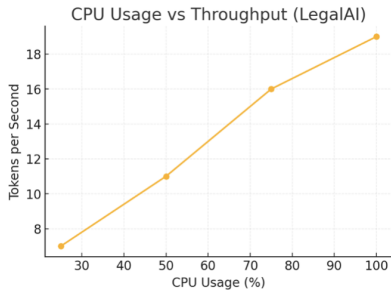


Fig. 9. CPU Throughput Comparison

J. Overall Performance Comparison

The radar chart provides a comprehensive multi-dimensional view of LegalAI's performance across all evaluation metrics, highlighting its strengths in privacy-preserving offline legal analysis.

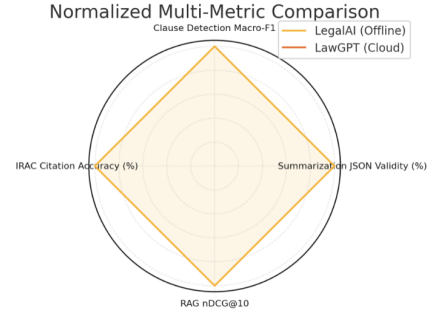


Fig. 10. Multi-Metric Radar Comparison

VI. CONCLUSION AND FUTURE ENHANCEMENTS

A. Conclusion

LegalAI is an entirely offline, privacy-preserving legal language model system to support document summarization, clause classification, and case-law reasoning through structured and citation-grounded results. The system was implemented based on a modular architecture that consists of a Vue.js-based frontend, Django REST Framework backend, LLaMA-based inference engine deployed locally, optional RAG, and secure local storage. Experimental testing proved that LegalAI produces competitive, and in many instances, superior results than cloud-based legal LLM systems. These significantly outperformed others in structured output validity, clause-level macro F1 accuracy, and citation reliability. When enabled, the retrieval pipeline significantly improved legal reasoning by situating responses in local legal sources. The design focuses on ethical and operational constraints important for legal practice, such as auditability, explainability, and data control. With fully local processing, strict citation rules, and structured outputs, LegalAI offers an alternative to the cloud AI tools for organizations that want confidentiality, regulatory compliance, and verifiable decision support.

B. Future Enhancements

Future iterations can incorporate hybrid symbolic neural retrieval integrating statute indexing, keyword search, and embedding vectors. Multi document context synthesis for cross jurisdictional reasoning and dynamic updating of embeddings when corpus materials change. Training on sector specific contracts, extraction of clause relationships and risk scoring based on contractual dependencies, and temporal clause analysis detecting version level changes. Future architectures should support edge cluster deployment for government or enterprise servers and distributed inference models allowing parallel execution of multiple legal queries. Offline connector for licensed legal databases, automatic citation cross

checking against statutory repositories, and version controlled law database management. Publicly distributable datasets that avoid restrictive licensing, standardised tests for citation pre-vision, IRAC compliance, and legal hallucination monitoring. Argument provenance tracing, highlighting every reasoning step and originating source text, counter argument generation to test case law reasoning robustness and tools for legal professionals to annotate, revise, or override AI reasoning in an audit trail format.

ACKNOWLEDGMENT

Not applicable.

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